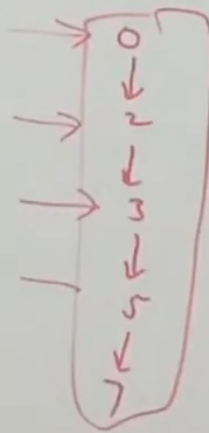
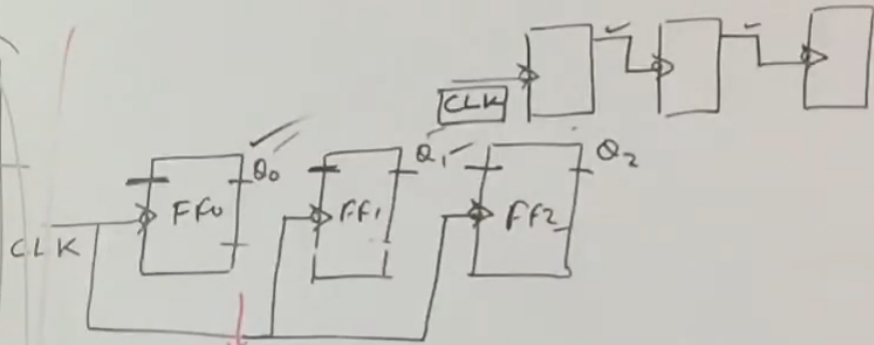


SYNCHRONOUS COUNTER DESIGN

Excitation Table

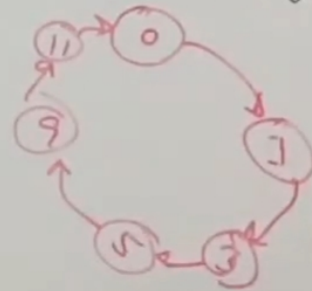
State	Next State	S	R	S	R
0	0	0	0	0	X
1	1	1	0	1	0
0	0	0	1	0	1
1	1	0	0	X	0

S	R	Q_{n+1}
<u>0</u>	<u>0</u>	<u>Q_n</u>
0	1	0
1	0	1
1	1	0



O/P $\rightarrow$ given

$I/P_s \rightarrow \text{Design}$



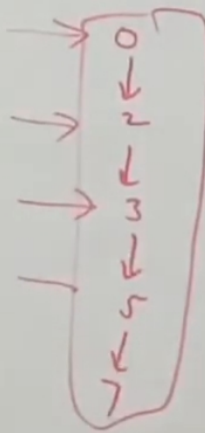
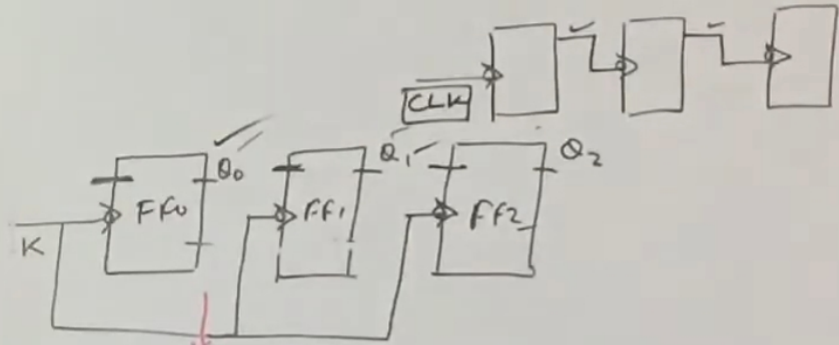
SYNCHRONOUS COUNTER DESIGN

0/ps

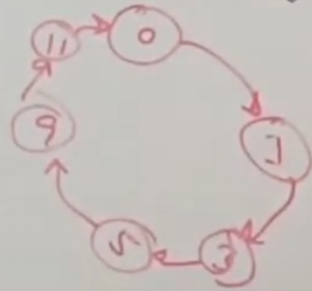
I/ps

PS	NS	J	K	J	K
Q_n	Q_{n+1}				
0	0	0	0	0	X
0	1	1	0	1	X
1	0	0	1	X	1
1	1	0	0	X	0

J	K	Q_{n+1}
0	0	$\underline{Q_n}$
0	1	0
1	0	1
1	1	$\bar{Q_n}$



$O/P \rightarrow \text{given}$
 $I/P_s \rightarrow \text{Design}$



SYNCHRONOUS COUNTER DESIGN

O/Ps		I/Ps	I/P
PS	NS	✓	D
Q_n	Q_{n+1}	T	
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	1

T	Q_{n+1}	D	Q_{n+1}
0	Q_n	0	0
1	$\bar{Q}_n$	1	1

